

Cyber hygiene: The concept, its measure, and its initial tests

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ABSTRACT

While policy makers to cyber security experts call for improving cyber hygiene, no one really knows what it means. In fact, there exists no scholarly research explicating the concept or its measurement. This research makes an important contribution by conceptualizing cyber hygiene, operationalizing it, empirically identifying its sub-dimensions, and developing an inventory for it. The research achieves this with a mixed-methods approach, where using a combination of experts and a convenience sample of Internet users, it develops the initial items reflecting the construct, empirically refines the items, confirms its dimensions, and validates its fit. The outcome is an 18-item Cyber Hygiene Inventory (CHI) that measures five distinct dimensions of user cyber hygiene. Finally, the research demonstrates why cyber hygiene matters. Using the CHI it shows how cyber hygiene significantly predicts aspects of human cyber interaction that are pivotal to cyber safety including user self-beliefs about technology, how they cognitively process information online, and their online banking behavior.

Following news of every major cyber breach, security experts all over the world recommend the need for improving cyber hygiene. This was the case after the Office of Personnel Management (OPM) hack that led to the exfiltration of 21.5 million highly sensitive records of current and former US federal government employees [1,2], after the infamous hack into the Democratic National Committee servers during the 2016 US presidential elections [3], and after the distributed ransomware attack that began by wiping out data within computer networks in UK's National Healthcare System (NHS) and subsequently spread to computing networks in over 100 countries [4].

And it's not just security experts, policy makers and governments are also calling for more cyber hygiene. For instance, responding to a Government Accountability Office (GAO) report detailing the continued use of Adobe Flash on federal government computers, in June 2018, Democratic Senator Wyden wrote a letter admonishing the National Security Agency (NSA), the Department of Homeland Security (DHS), and the White House for their failure to practice cyber hygiene [5]. Similar criticisms, for not doing enough to ensure better cyber hygiene practices, have also been levelled against President Obama and his teams [6].

But while many experts lament the lack of cyber hygiene, it's unclear what cyber hygiene really means. In fact, there isn't a single

academic research article that explicates the construct or presents its measurement. Adding to the confusion, a search on Google's search engine returns hundreds of web pages authored by bloggers, IT companies, and different shades of cyber security experts espousing all manner of suggestions from the obvious (e.g., safeguard your emails better) to the highly specific (e.g., ensure you use a certain specific operating system or anti-virus software) [7,8]. The online advice also runs the gamut of approaches. Some call for improved vigilance, others for heightened awareness, and some others for users to completely change their behaviors (such as calling a bank on the phone instead of responding to any emails from them). Further muddying the waters are varying guidelines from different organizations. For instance, recently the National Institute of Standards and Technology (NIST) published guidelines where they discussed cyber hygiene in the context of IT patch management practices [9]; the European Union recently announced a program aimed at improving cyber hygiene among children where their focus was on preventing cyber bullying and the spread of fake news [10]; while the Monetary Authority of Singapore (MAS) included multi-factor authentication, the use of firewalls, and anti-virus programs in their recent publication [11].

What's clear from all this is that there is not a singular action that comprises cyber hygiene. In actuality, it appears to mean different

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things—from effective patch management to knowing not to use certain types of software to backing up computers to avoiding cyber bullying and detecting fake news—and applies to different people, from CSOs and IT managers to everyday Internet users, children, and even the president of the United States. Besides, it's unclear whether these activities even matter and, if they do, how. Having such an important concept be so vaguely defined, broadly construed, and, for the most part, untested, does little to improve cyber resilience. All it does is confuse IT managers and end-users by providing conflicting advice that further weakens our ability to protect the cyber space. If we ever hope to achieve cyber resilience, we need to clearly explicate what cyber hygiene means and why it matters.

This begins by defining the meaning of cyber hygiene and locating its boundaries. Once defined, we need to develop measures for it, which then permit us to empirically measure the construct, quantify cyber hygiene, identify its dimensional structure, and assess the scope of its impact. These two steps are pivotal in order to assess the present levels of cyber hygiene, i.e., know how much of it different users possess, identify what's lacking, and find effective ways to mitigate those weaknesses. Thus, defining and developing measures for cyber hygiene is not just important but necessary for achieving cyber resilience.

The current paper outlines a series of research studies that achieve all of this. It begins with a conceptual definition of cyber hygiene as it relates to general Internet users—the largest group of online consumers, who are the primary target and the main conduit through which hackers enter into organizational networks [12]. Next, using a series of empirical studies, the research locates the dimensions of cyber hygiene, develops measures for each dimension, and refines the scale using a series of research studies.

To develop the initial measures, the research applies the Concept Mapping approach developed by Trochim [13], a mixed-method approach widely applied in different fields for developing multi-dimensional measures of complex phenomenon [14]. Next, the research uses data from a survey of adult Internet users to ascertain the scale's dimensionality and fit using a Confirmatory Factor Analysis (CFA) approach. Following the construct validation steps outlined by MacKenzie, Podsakoff, and Podsakoff [15], the research then establishes the convergent, discriminant, and concurrent comprising each dimension of cyber hygiene. The final outcome of the paper is a conceptual definition and a validated, multidimensional, quantitative measure for the cyber hygiene of Internet users called the Cyber Hygiene Inventory (CHI). Then using the measure, the study assesses the impact of the CHI on facets of human cyber interaction that are important for cyber safety—thereby demonstrating the real impacts of cyber hygiene.

The paper begins with the conceptual definition in the next section followed by the concept mapping approach used in Study 1 and the results. This is followed by a presentation of Study 2, its survey approach, the data, analysis, and the results of the research, followed by a discussion of the findings, some caveats about the research approach, and suggestions for future research.

1. Defining cyber hygiene

Cyber hygiene draws from the concept of personal hygiene from the public health literature. In a major report examining cyber hygiene practices across various nations, the European Union Agency for Network and Information Security (ENISA) stated that “cyber hygiene should be viewed in the same manner as personal hygiene and, once properly integrated into an organization will be simple daily routines, good behaviors and occasional checkups to make sure the organizations online health is in optimum condition.” (ENISA, December 2016, pp. 4) [38]. But the report also found that there was no consensus on what cyber hygiene meant or even entailed. Various nations including the United States had their own practice recommendations and advisories and there is no single standard or commonly agreed upon mechanism for an organization to assess, evaluate or demonstrate cyber hygiene.

The report underscores the need for the present study but also provides little additional guidance for the development of the concept.

Outside of this, there have been no attempts to develop a measure for cyber hygiene. The extant research on the social science on cyber security has instead focused on end-user security behaviors and their predictors. Stanton et al. [16], for instance, developed a taxonomy of password-related behaviors and found two predictive dimensions, the users' technical knowledge needed to change passwords and their intentionality. Likewise, Ng, Kankanhalli, and Xu [17] focused on email security related behaviors and found they were predicted by various end user attitudes and self-beliefs. Finally, Egelman and Peer [18] developed a four-dimensional Security Behavior Intentions Scale (SeBIS) that assesses end-users' compliance with computer security advice.

All these approaches focus on measuring the frequency of enactment of one or a few cyber safety behaviors by implicitly assuming that users are aware of all these behaviors. This is both a conceptual and operational limitation. Conceptually, the focus tends to be limited to a few behaviors that are a priori determined by the researcher and whose availability in the end users' cognition (in terms of awareness or understanding) is assumed to be implicit. In consequence, the empirical approach tends to be limited to a few simple cyber safety actions that are expected to be widely known to users.

The present research instead focuses on the entirety of possible cyber hygiene actions that are empirically identified and whose relationships are located by security experts and then validated by end users. Lacking research that directly addresses the concept of cyber hygiene, the research looked for guidance in the source of the term cyber hygiene—the concept hygiene.

The root of the word “hygiene” can be traced back to ancient Greece and according to the World Health Organization, it encompasses the practices of personal cleanliness that prevent the spread of disease-causing organisms [19]. It includes different types of hygiene, from sanitary hygiene practices such as washing hands and bathing, to dental or oral hygiene, sleep hygiene, food hygiene, mental health hygiene, sexual hygiene, and others.

Three aspects of how hygiene is construed in the public health literature are noteworthy. First, hygiene is a multidimensional concept, with many underlying facets governing the various sub elements that altogether comprise a person's hygiene. Second, hygiene is usually a guideline, where the focus is on what someone should do and should be aware of doing. For instance, washing hands is a blanket action that many people are expected to be aware of and to engage in, regardless of whether people understand the process of disease transmission. Third, because most hygiene practices are rooted in contextual and cultural realities, hygiene is usually broadly defined. For instance, people working in healthcare settings might wash their hands more frequently, even using harsher chemicals and disinfestations, while people who don't work in such settings might simply do it a few times using soap and water. There are also cultural differences in hygiene practices. In cultures where people eat food with their hands, washing hands prior to eating even at restaurants is considered a sanitary practice, something most people in the US or Europe seldom do. Thus, washing hands regularly with soap and clean water is a broadly prescribed hygiene practice that doesn't specify the frequency, timing, or the type of soap that should be utilized. This allows for the broad guidelines to be easily disseminated and contextualized based on whether it is being applied within the setting of a home, in a restaurant or food service location, in a healthcare setting, or within a specific culture.

These three facets of hygiene translate to our conceptualization of user cyber hygiene as “the cyber security practices that online consumers should engage in to protect the safety and integrity of their personal information on their Internet enabled devices from being compromised in a cyber-attack.”

In-line with how hygiene in the public health realm is defined, this definition of cyber hygiene provides a first-order conceptualization—an overarching theoretical definition that does not specify or constrain the

constituent factors that form the sub dimensions of cyber hygiene. This way the sub dimensions can be determined empirically, as is done in the current paper. The theoretical conceptualization also allows the various sub facets of cyber hygiene to be determined by the context in which it is being applied. Thus, similar to the guidelines on personal hygiene, the cyber hygiene definition and guidelines can be broad enough to apply to the general consumers yet moldable to specific professional or cultural contexts. For instance, for individuals working in organizations, the definition of cyber hygiene would extend to include the protection of information and devices entrusted to them and under their purview from being compromised. The subdimensions that comprise cyber hygiene for employees would then be determined by the role of the employee, the type of the organization, or the sector within which the organization operates—all of which could influence some of the cyber hygiene actions the employee is expected to be aware of.

Not only does this approach accommodate a wide range of user types, but it is also flexible in that it accommodates changes in the technological environment overtime. For instance, if at some future date, quantum computing was to necessitate a new form of password management practice, these practices could be easily integrated into the existing measurement model without needing to change our fundamental conceptualization of cyber hygiene. Finally, other than contextualizing hygiene differently based on whether it is enacted within organizations or at home by consumers, the conceptual definition does not specify a device, operating system, or user environment, making the definition scalable and broadly applicable across technological products and a range of use cases. Thus, the conceptual definition accommodates a wide range of users, technologies, related practices, and changes overtime.

2. Operational definition of cyber hygiene

When it comes to personal hygiene, individuals do not need to understand disease transmission in order to enact it. For instance, people do not need to know how disease transmission occurs in order to know to cover their mouths when sneezing. This is because understanding the specifics of how various diseases are transmitted is far too complex and unnecessary for most users who need to simply be aware of the need to enact the hygiene practice.

Furthermore, from a public health point of view, individual do not always need to enact or demonstrate the performance of the behavior to a researcher but indicate their awareness of the behavior. This is because many actions are private (e.g., people cannot demonstrate sexual hygiene), some are personal-need driven (e.g., only people who wear contact lenses remove them prior to going to bed), while others are only enacted within certain contexts (e.g., using a tissue when sneezing; [20]). Hence, when it comes to personal hygiene, research and public health interventions usually focus on building awareness and it is the public's level of awareness that is the focal measure of interventions and assessments [21]. Following this, at the operational level, the focus of cyber hygiene measurement is also on the level of awareness of various cyber safety actions.

In addition to being consistent with how hygiene is generally measured, focusing on awareness levels of various cyber hygiene actions has three distinct advantages. For one, awareness always precedes behavior, which is why even most public health campaigns focus on improving it before engaging in behavioral change interventions [21]. Thus, measuring awareness allows us to get to the root of the problem of cyber hygiene. Two, the frequency of cyber behaviors that people engage in is contingent on the device, the application, the service, or the operating system the user is interacting with. For instance, users might alter their privacy settings on social media platforms each time they post information online because the platform allows for it and makes such changes easy, while they might change the privacy settings on a new email service perhaps once, usually during initial setup. Thus, focusing on awareness makes it possible to cluster actions based on how

users cognitively or internally consider or relate to these actions rather than the frequency with which each of these actions are performed that is contingent on factors outside the users' control (e.g., the device, software, or application). Finally, measuring behavioral enactment ideally requires multiple response scales for capturing each aspect of the behavior's evaluation, potency, and level of activity. Instead, focusing on awareness necessitates a single response scale, which allows for fewer overall questions and the use of a simpler measurement process.

Given these advantages, the research operationally focused on developing a measure of the awareness of different cyber safety actions that fit the conceptual definition of cyber hygiene. Similar to personal hygiene, the expectation is that cyber hygiene is a multidimensional construct. The dimensions and their individual measures were, therefore, derived empirically using the Concept Mapping approach, presented in the next section.

3. Concept mapping

Concept mapping is a mixed-method, statistical approach for measuring and understanding complex, multi-faceted phenomena [13]. The methodology has been applied in business, urban planning, and healthcare research for visualizing the conceptual relationship between different constructs, empirically developing their measures, and studying complex phenomenon [13,14,22,23]. The overall approach borrows from the traditional Likert scale development approach for the development of items lists, but allows for user participation in the determination of the concept names and structure, thereby resulting in concept arrays with high ecological validity. This makes the approach particularly well-suited for studying multi-faceted and technical phenomenon.

Concept mapping also accommodates the five-step procedure for scale development outlined by MacKenzie et al. [15]. Their five-step procedure for scale development, visually presented in Fig. 1, begins with conceptualizing a construct, followed by the development of measures, model specification, scale evaluation and refinement, validation.¹ Concept-mapping follows five steps, which altogether accomplish these. These five steps, visually presented in Fig. 1, begin with the development of a concept definition or focus prompt which is used for brainstorming candidate statements in the second step. This is followed by a refining process in the third step where items are sorted by a small number of experts who independently organize the items into groups, and give it a descriptive name. The first three steps require a small pool of domain experts and past research has accomplished reliable estimates using anywhere from 5 to 7 experts [24].

The sorted items are then organized into a binary symmetric matrix in the fourth step, and subject to a two-dimensional multidimensional (MDS) analysis. The X-Y value for each item defines its position in the map and items that are closer to one another are considered to represent similar constructs than those that are further away. Following this, in the last step, the MDS solution is subjected to a hierarchical cluster analysis using Ward's algorithm. The descriptive names given by each expert during the sorting task is used to define the cluster; the clusters and the maps are reviewed by the experts who collaboratively determine the components and definitions of each cluster.

The final scale development procedure, validation, as outlined by MacKenzie et al. [15], is achieved using a separate sample of adult Internet users. In the sections below, the paper outlines the outcomes of the Concept Mapping process and how the scale was validated.

¹ A sixth step, norm development, where a scale's ideal values are prescribed, requires a scale's deployments across different samples and populations. This, given the preliminary nature of the CHI, is excluded here and a topic for future research.

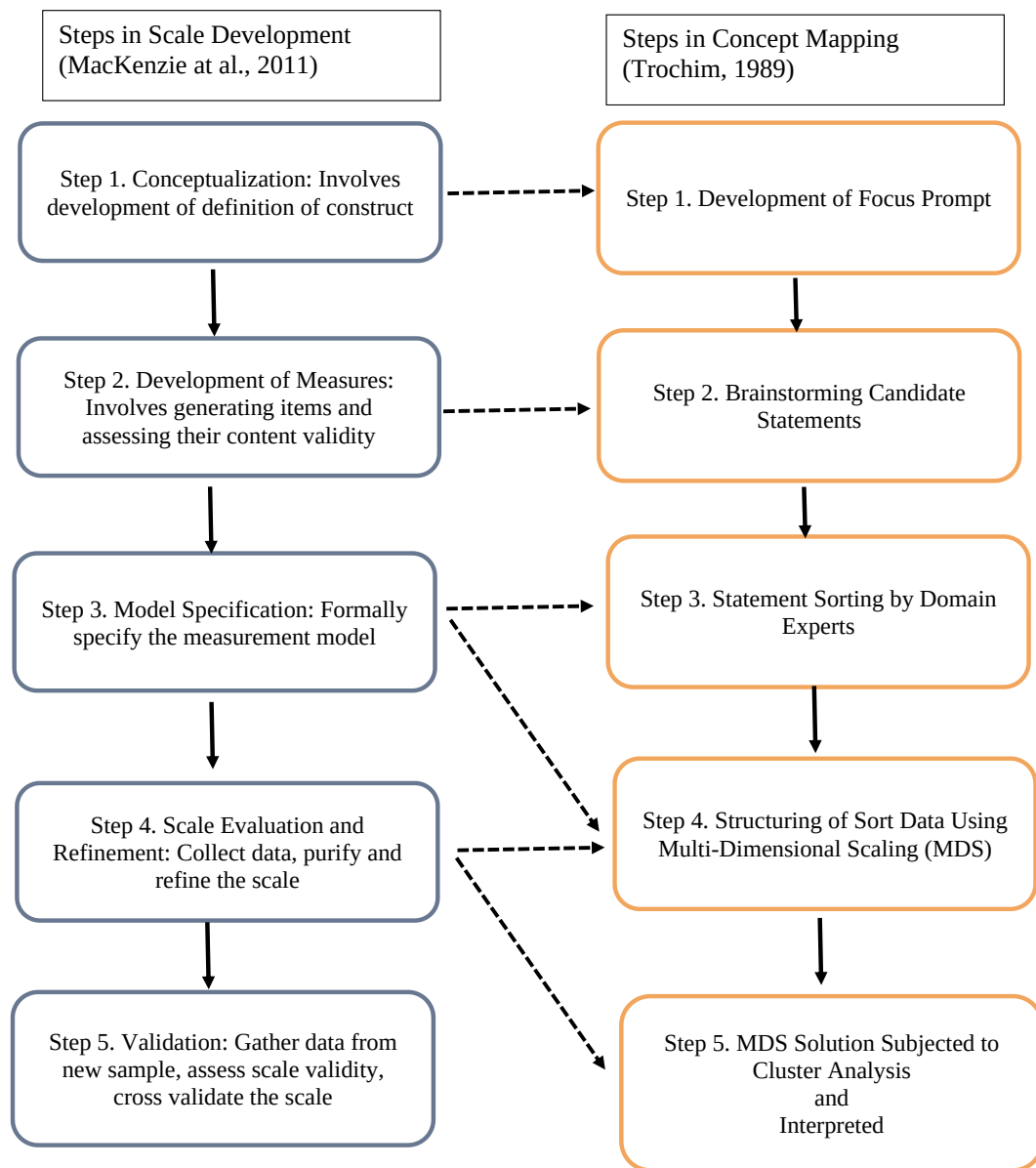


Fig. 1. Procedures for scale development outlined by MacKenzie et al. [15] displayed on the left and the Concept Mapping process displayed on the right, with dotted lines indicating how Concept Mapping accommodates each scale development procedure. Some facets of scale refinement and scale validation was, as outlined by MacKenzie et al., achieved using a separate sample.

3.1. Initial item pool generation

Using the conceptual definition of cyber hygiene as a focus prompt, the research began by deriving the initial pool of items reflecting the definition. For this, the study enlisted the help of half a dozen IT professionals drawn from the university CSO's office, experts working on cyber security employed in the US federal government, and from a Silicon Valley technology firm. Through discussions and interviews, a list of 71 candidate items was initially generated.

The items were then independently assessed by members of the research team. Each expert evaluated the items on two facets of translation validity: face validity, or whether, when read, on the surface-level each item appeared to be a good translation of the construct; and content validity or whether the content of each item appropriately fit the operationalization. This process led to the removal of some poorly worded items, combining of duplicate items, rewording of some items that were inaccurate or unclear, and the development of simple definitions for technical terminologies (such as what is a VPN, SSL, two-

factor authentication, etc.) that some of the evaluators felt would be helpful to end-users when responding to the inventory. The process netted a refined list of 39 items that all the experts felt best represented the conceptual definition of cyber hygiene.

3.2. Sorting and multidimensional scaling

The next step in Concept Mapping involves sorting. While the task can be accomplished by a handful of participants [24], the present study utilized 36 researchers and psychologists working in cyber security from Singapore's Home Team Behavioral Science Center (HTBSC), which operates within the security services sector in South-east Asia, all of who had cyber security expertise. Each expert independently sorted the statements into non-overlapping groups and provided a descriptive name for each group. The reliability of the sorting procedure was checked using the average split-half matrix reliability assessment procedure suggested by Trochim [25]. The analysis netted a reliability of 0.78, close to the statistic obtained in prior

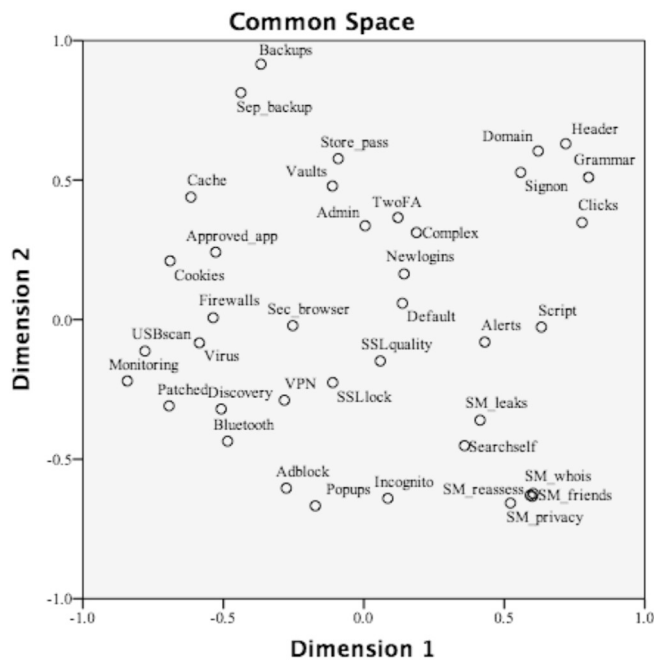


Fig. 2. Two-dimensional MDS solution for items sorted by the expert sample; full items/statements corresponding to each item title is provided in Table 1.

research by Trochim. The sorted data was then subjected to a two-dimensional non-metric MDS algorithm, the output of which is presented in Fig. 2.

The MDS solution was next subjected to a cluster analysis using Ward's algorithm and a Scree plot of the coefficients from the agglomeration schedule (which shows how the algorithm progressively combined cases to form clusters) was used to identify the most parsimonious number of clusters [26]. The plot identified a five-cluster solution, which is presented in Fig. 3.

Next, each of the five clusters was collaboratively accorded a descriptive name. This was done by using the descriptors provided by the experts during the sort task and through discussions with the research team to identify the best descriptor for each group of items. Whenever descriptors were in conflict, the items within each cluster were individually assessed, and the descriptor that best represented the majority of items in each cluster was selected. Using this process, the research team settled on dimension names that conveniently fit the acronym: SAFETY, where S: storage and device hygiene; A: authentication and credential hygiene; F: Facebook and social media hygiene; E: email and messaging hygiene, T: transmission and browsing hygiene. The final character Y: your or yourself, was added to create a mnemonic and to underscore the responsibility of the individual user in achieving cyber hygiene. Table 1 lists the items that comprise each cluster.

3.3. Scale validation

Having developed the initial items that fit the Cyber Hygiene Inventory (CHI), the research next set out to validate the scale. The focus of scale validation is threefold: a) establish the dimensionality and factorial validity of the scale, b) demonstrate the convergent and discriminant validity of the scale, and c) assess the predictive and concurrent validity of the scale (i.e., show that the measures predict what they should and distinguish between groups or individuals who demonstrate more or less of the attributes the scale should theoretically distinguish between). Analytically, the focus of this process is on determining the internal coherence of the item to factor relationship, and not on demonstrating the external validity or generalizability of the scale measures. While the latter requires a representative sample, the former can be achieved using a convenience sample. The research team, thus, acquired validation data using a snowball sampling process. Rather than focuses solely on college age students, the research endeavored to include a broad segment of Internet users who ranged in age and occupation status.

To this end, the research team used Google Forms to convert the cyber hygiene items into an online survey. The survey included all 39 items from the CHI along with additional measures for scale validation

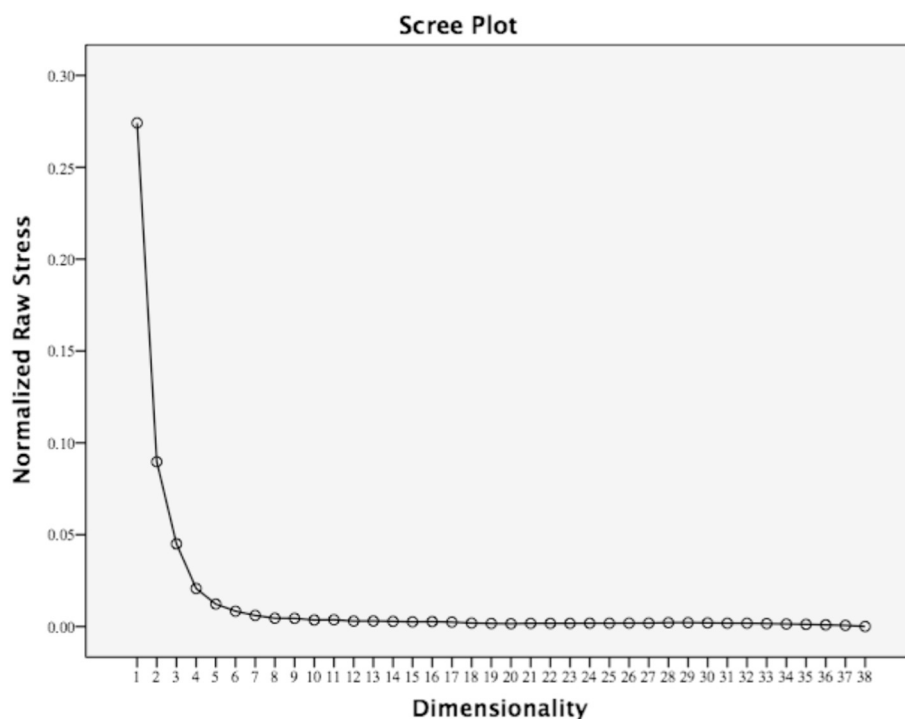


Fig. 3. Scree plot of the coefficients from the agglomeration schedule produced by Ward's Cluster Analysis.

Table 1

List of items that comprise each cluster, along with item titles (used in Figs. 2 and 4) and list of final 18 retained items that comprise the Cyber Hygiene Inventory (CHI).

	Original	Codes used	Original items in each cluster/ factor	Final items retained after CFA
1	Using a VPN when in an open/public Wi-fi network	VPN	F2	
2	Checking the lock icon (also known as SSL) on the web browser to see if a website uses encryption	SSLlock	F2	
3	Checking the quality of the SSL certificate when doing online financial transactions	SSLquality	F2	F2
4	Restricting who can connect to your device via Bluetooth	Bluetooth	F1	
5	Managing computer discovery on public networks	Discovery	F2	
6	Enabling firewalls on your computing devices	Firewalls	F1	F1
7	Using script management tools (such as Ghostery or NoScript)	Script	F2	
8	Using ad-blocking tools such as AdBlock	Adblock	F1	
9	Clearing cookies on browser	Cookies	F1	
10	Clearing browser cache	Cache	F1	
11	Disabling pop-ups on browser	Popups	F1	
12	Using secure browsers	Sec_browser	F4	
13	Running a virus scan on any new USB or external storage device	USBscan	F1	F1
14	Maintaining a backup of all my important files on a cloud server	Backups	F1	
15	Creating a separate backup of all your files on a backup disc separate from your network	Sep_backup	F5	
16	Checking an incoming email's header	Header	F5	F5
17	Checking a sender's email domain name	Domain	F5	F5
18	Clicking on hyperlinks or attachments in email from unknown senders	Clicks	F5	
19	Checking to see if email requests have grammatical or typographical errors	Grammar	F5	F5
20	Monitoring when and where you last signed onto your email account	Signon	F1	
21	Checking your device to ensure it has the latest OS, software update, or patch	Patched	F1	
22	Downloading approved software from the operating system's app market	Approved_app	F3	
23	Monitoring different processes such as CPU, power, or network usage on your device	Monitoring	F1	F1
24	Changing default username from administrator to something unique on all Internet enabled devices	Admin	F4	F4
25	Changing default passwords on all Internet enabled devices	Default	F4	F4
26	Keeping virus protection updated	Virus	F1	F1
27	Managing how your browser stores passwords	Store_pass	F4	F4
28	Creating complex logins and passwords	Complex	F4	
29	Creating new/unique logins and passwords for all your online sign-ins	Newlogins	F4	F4
30	Enabling two-factor or multi-factor authentication for logins	2FA	F4	
31	Storing logins and passwords on encrypted online password vaults	Vaults	F2	F2
32	Using Incognito or Private Mode when routinely surfing the Internet	Incognito	F3	
33	Searching your name online on a search engine	Searchself	Dropped	
34	Placing online alerts from your name or personal information	Alerts	F2	F2
35	Assessing the authenticity of social media friend/information requests	SM_friends	F3	F3
36	Knowing who you are connected to on social media	SM_whois	F3	F3
37	Reassessing social media friends/connections	SM_reassess	F3	F3
38	Managing privacy settings on social media platforms	SM_privacy	F3	
39	Ensuring the location information is not leaked in posts	SM_leaks	F3	F3

Note: F1 = Storage and device hygiene, F2 = Transmission hygiene, F3 = Facebook and social media hygiene, F4 = Authentication and credential hygiene, F5 = Email and messaging hygiene.

(detailed in sections below). So as to improve comprehension, wherever necessary, technical terms were clarified with the definitions developed by the research team. After procuring human subjects research clearance, the survey link was shared by the research team with others in their personal and professional network. Multiple interns, researchers, and staff members from HTBSC also shared the link with universities, schools, colleges, and other organizations who were invited to complete the survey and solicit participation from others in their network.

This overall process lasted for four-weeks from June–July 2017, at the end of which the collected data were examined to ensure that only fully completed surveys were retained. The final dataset consisted of 404 completed surveys of Internet users.

3.4. Sample demographics

The median age of the respondent was 26 years ($SD = 11$) and 54% of them were males. The sample had an average monthly household income of SGD8500 ($SD = SGD4500$); the median education level of the respondent was a GCE-A level (equivalent to a high school diploma); 69% of the sample were students, 25% were working professionals, and the remainder were a mix of self-employed individuals, retirees, and homemakers. Closely paralleling the census based racial composition of Singapore, 71% of the respondents were Chinese, 16% were Malay, 9%

were Indian, and the remaining 3% were of mixed ethnicity [27].

3.5. Confirmatory factor analysis

To assess factorial validity, the study conducted a Confirmatory Factor Analysis (CFA) for the five latent constructs identified by the Concept Mapping process. As is usually the case, the initial analysis achieved a less than satisfactory fit. Using the modification indices, items that led to the poor fit were identified and dropped. From the Storage and device integrity hygiene factor, 8 items from the original 12 were dropped (items dropped: item 2, item 3, item 5, item 10, item 11, item 18, item 19, and item 20); from the Authentication and credential hygiene factor's original 7 items, 3 were dropped (item 21, item 23, and item 30); from the Facebook and social media hygiene factor's original 7, 3 items were dropped (item 17, item 37, and item 39); from the Email and messaging hygiene factor's original 5, 2 items (item 4 and item 9); and from the Transmission and browsing hygiene factor's original 7, 4 items were dropped (item 1, item 15, item 28, and item 33.) After dropping these items, the CFA produced a good fit: $\chi^2(122) = 283.92$, CFI = 0.97, TLI = 0.96, SRMR = 0.04, and RMSEA = 0.06 (90% CI = 0.05, 0.07). Table 1 presents the final list of items that was retained based on the CFA. Fig. 4 presents the CFA results and Table 2 presents the model fit indices.

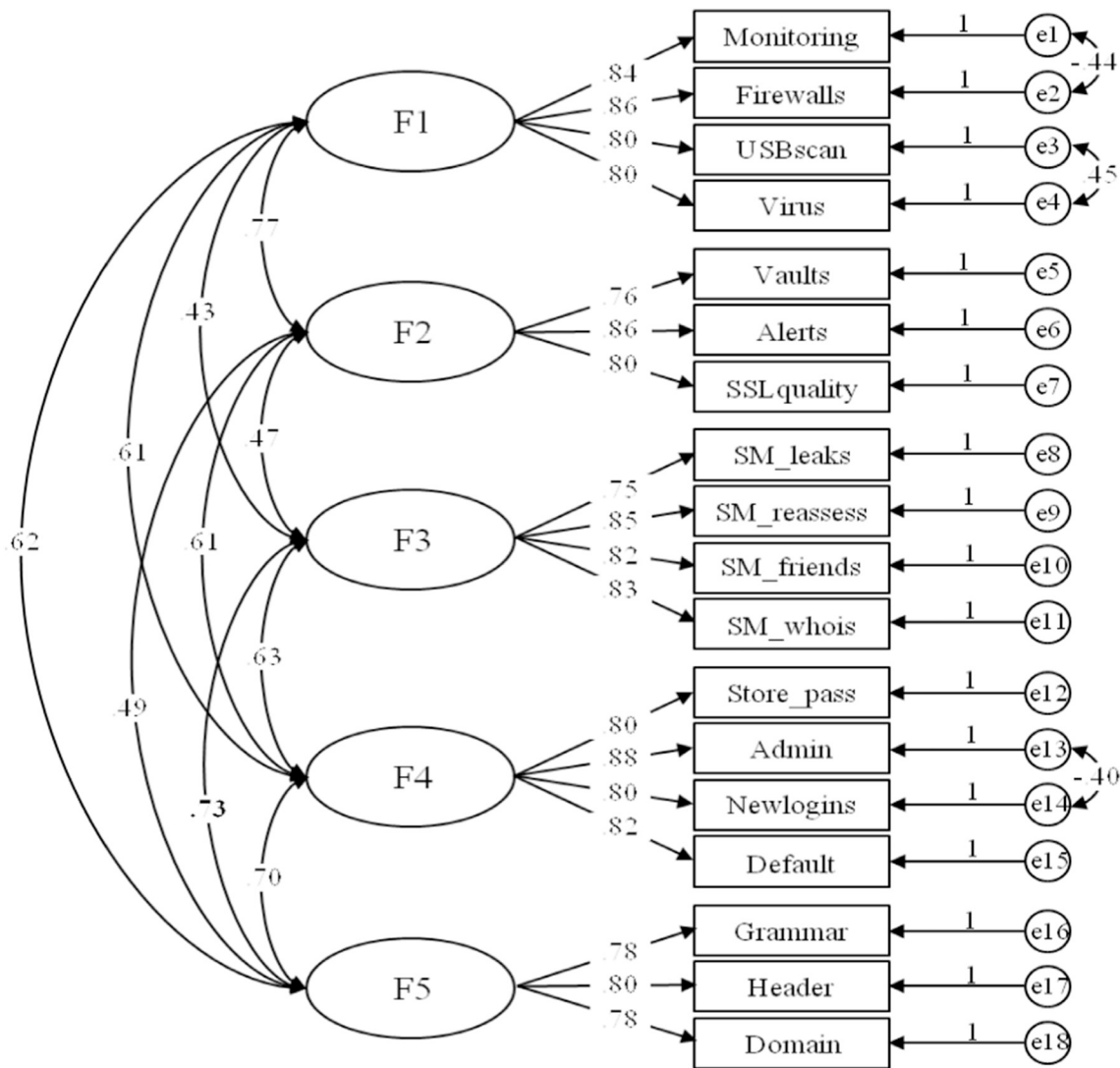


Fig. 4. Confirmatory factor analysis (CFA) for the awareness variables; standardized estimates provided next to each path; factor names are as follows: F1 = Storage and device hygiene, F2 = Transmission hygiene, F3 = Facebook and social media hygiene, F4 = Authentication and credential hygiene, F5 = Email and messaging hygiene.

3.6. Convergent and discriminant

Next, the analysis examined the convergent validity for the five constructs using Hair, Black, Babin, Anderson, and Tatham's [28] formulas for the composite reliability (CR)² and the average variance extracted (AVE).³ Constructs have convergent validity when the composite reliability exceeds the criterion of 0.70 and the average variance

² According to Hair et al. [28], the composite reliability is calculated by the following formula,

$$CR = \frac{(\sum_{i=1}^n \lambda_i)^2}{(\sum_{i=1}^n \lambda_i)^2 + (\sum_{i=1}^n \delta_i)}$$

where λ is the factor loading and δ is the error variance.

³ According to Hair et al. [28], the average variance extracted (AVE) is calculated by the following formula,

$$AVE = \frac{\sum_{i=1}^n \lambda_i^2}{n}$$

where λ is the factor loading and n is the number of observed variables.

Table 2

Model fit indices for confirmatory factor analysis (awareness variables).

χ^2	df	p	GFI	SRMR	TLI	CFI	NFI	RMSEA
283.92	122	0.000	0.930	0.046	0.957	0.966	0.942	0.057

extracted is above 0.50 [28]. Table 3 shows that all CRs exceeded the criterion of 0.70 and all AVEs were above 0.50. Therefore, convergent validity was supported.

Discriminant validity is assessed by comparing the AVE to the squared correlation estimate between factors. Hair et al. [28] asserted that if the AVE estimates for each construct are greater than the squared correlation estimate for the constructs, the discriminant validity is supported. As shown in Table 4, all AVEs (diagonal cells) are larger than the squared correlations between constructs, confirming discriminant validity.

3.7. Concurrent validity

Concurrent validity is a type of predictive validity that assesses whether a measure can distinguish between groups it should

Table 3
Convergent validity (awareness variables).

CHI dimension	Composite reliability	Average variance extracted (AVE)
F1: Storage and device hygiene	0.855	0.679
F2: Transmission hygiene	0.761	0.652
F3: Facebook and social media hygiene	0.856	0.660
F4: Authentication and credential hygiene	0.853	0.677
F5: Email and messaging hygiene	0.757	0.622

Table 4
Discriminant validity result (awareness variables).

CHI dimension	F1	F2	F3	F4	F5
F1: Storage and device hygiene	0.679				
F2: Transmission hygiene	0.594	0.652			
F3: Facebook and social media hygiene	0.182	0.222	0.660		
F4: Authentication and credential hygiene	0.371	0.367	0.391	0.677	
F5: Email and messaging hygiene	0.382	0.243	0.530	0.486	0.622

Note. Average variances extracted (AVE) estimates are reported on the diagonal. Squared correlation estimate between factors is reported below the diagonal.

theoretically distinguish between. For instance, a measure for job satisfaction should distinguish between employees who are likely to remain in the company longer versus those who are likely to find employment elsewhere [29]. Following this, the research examined concurrent validity by ascertaining whether individuals who had high awareness were significantly likely to engage in more of those behaviors.

In the web survey, the frequency of enactment of each CHI item was measured using a 1–5, never to always scale where respondents were asked to indicate how often they engaged in each of the behaviors. Using the CFA results, items were summated and averaged resulting in five cyber hygiene enactment subdimensions each corresponding to an awareness dimension. The frequency with which users engaged in actions for the five latent constructs were as follows: Storage and device hygiene ($M = 2.67$, $SD = 0.98$, α reliability = 0.86); Transmission and browsing hygiene ($M = 2.16$, $SD = 0.95$, α reliability = 0.75); Facebook and social media hygiene ($M = 3.26$, $SD = 1.05$, α reliability = 0.89); Authentication and credential hygiene ($M = 2.90$, $SD = 1.03$, α reliability = 0.84); and Email and messaging hygiene ($M = 3.28$, $SD = 1.12$, α reliability = 0.84).

Using the mean of each enactment, the research created a low frequency and high enactment group. Independent sample *t*-tests were then used to compare whether the high versus low enactors were significantly different in their levels of awareness.

Table 5 presents the analysis. For each dimension of cyber hygiene enactment, those who were in high enactment group had significantly higher awareness than those in the lower enactment group. Thus, because enactment is predicated on awareness, results point to cyber hygiene awareness discriminated between those who enacted more of the behavior than those who did less of it.

3.8. Why does cyber hygiene matter?

Having validated the CHI, the final consideration is one of value. While many have implicitly assumed cyber hygiene matters, how it does has never been explored, partly because there has been no clear conceptualization of the construct, and because there has been no measure for it. The CHI provides the first opportunity to do just this.

To this end, the survey focused on three different constructs related to cyber hygiene, each tapping into a pivotal facet of the human-cyber interaction process: a technology related user belief, a cognitive processing factor, and an online behavior that is of particular importance to cyber security research.

The first construct was technological self-efficacy, defined as the users' confidence in their ability to deal with issues related to technology [30]. Self-efficacy is often assessed by researchers in their quest to understand Internet user behavior (e.g.; [31,32]). Theoretically, one would expect users with high levels of cyber hygiene, because it translates to higher levels of perceived awareness of different forms of cyber safety action, to also feel more confident in their ability to deal with technology issues directly related to cyber security. The research measured cyber security self-efficacy using a single question that directly asked respondents to rate their overall confidence in their ability to detect a malware infection on their computer or device using a 1–5, “Not at all confident” to “Very confident” response scale ($M = 2.81$, $SD = 1.04$).

The second construct was the extent to which users engaged in systematic processing when online [33]. This construct is based on the Heuristic-Systematic Model (HSM) [34,35], one of the most widely applied dual-processing theories of cognitive processing. This theory posits that individuals cognitively process information either using mental shortcuts called heuristics or by systematically connecting information from the context to knowledge repositories stored in your memory. From decision-making quality point of view, systematic processing is the preferred mechanism of processing because it is derived from prior knowledge, which when it is of good quality, leads to better outcomes.

In the domain of cyber security, prior research by Vishwanath, Harrison, and Ng [36] has empirically demonstrated that individuals who systematically process information are significantly less likely to fall victim to spear phishing. Much of today's cybersecurity education and training programs, therefore, aim to improve the quality of this form of processing. From the current research point of view, the expectation is that individuals who possess high cyber hygiene, because they have higher awareness of different cyber safety routines, would be more significantly more likely to systematically process information.

While systematic processing is often measured on a contextual basis, usually in reaction to a particular stimulus, research by Griffin, Neuwirth, Giese, and Dunwoody [37] has also measured it using general population surveys where respondents were asked to indicate their overall propensity to systematically process information about certain environmental risk events. Following this, the current study included a 4-item measuring the extent to which respondents systematically processed spear phishing emails. The measure originally, drawn from prior research by Vishwanath et al. [36], had sample items that read “In general, when you receive an email, you spend time thinking about what the email is asking you to do,” “...you draw links as to what the email is asking,” and so on. Items were scaled from 1 to 5, Strongly Disagree–Strongly Agree, with the overall scale netting a mean = 3.78 ($SD = 0.76$; α reliability = 0.71).⁴

After assessing user self-beliefs and cognitive processing, the research chose a behavior that is of particular concern for cyber security researchers and practitioners: online shopping. Online shopping websites are by far the most utilized commercial properties on the Internet. An estimated 1.66 billion people shopped online in 2017, with the Asia-

⁴ The alpha reliability of heuristic processing closely matched Griffin et al.'s [37] who netted an omega of 0.68 for the scale in their study.

Table 5*t*-test comparing cyber hygiene awareness measured using the CHI against cyber hygiene enactment.

CHI dimension	High enactment group's awareness levels <i>Mean</i> (<i>SD</i>)	Low enactment group's awareness levels <i>Mean</i> (<i>SD</i>)	<i>t</i> -test results
F1: Storage and device hygiene	3.64 (0.89)	2.82 (1.03)	$t(402) = 8.47, p < .001$
F2: Transmission hygiene	3.38 (0.99)	2.30 (1.06)	$t(402) = 10.24, p < .001$
F3: Facebook and social media hygiene	4.33 (0.62)	3.19 (0.92)	$t(402) = 14.53, p < .001$
F4: Authentication and credential hygiene	4.09 (0.79)	3.01 (0.98)	$t(402) = 11.99, p < .001$
F5: Email and messaging hygiene	4.08 (0.82)	3.03 (1.06)	$t(402) = 11.10, p < .001$

Pacific region accounting for 12% of global online retail sales in 2016.⁵ Online shopping websites all over the world are routinely targeted through phishing and spoofing (creating fake copies of the site) attacks and many offer their customers broad guidelines on detecting a fake website. Using some of these as a basis, the research team identified eight cyber safety actions that most users are expected to engage when ascertaining the veracity of an online shopping website. Sample items assessed whether the respondent verified if “the site was using an SSL/lock icon on the browser next to the URL,” “whether the site was asking for bank pin numbers or other financial information not necessary for completing transaction,” “the seller was sending emails with shipping information when you haven't purchase products,” “whether the site appeared to not list any contact information,” “whether the site was using a domain name that appears similar to other popular website,” “whether the layout of the page was not consistent with what you'd expect from that type of shopping site,” “the quality of images, colors, fonts... appears to be of lower quality,” and “the quality of images... appear to be inconsistent with what you'd expect from that type of shopping site.” Responses were coded as 1 = yes and 0 = no, with the overall responses, which ranged from 0 to 8, cumulated ($M = 5.38$, $SD = 2.03$).

Altogether, the three constructs encompassed the belief, processing and behavioral facets of human cyber interaction. The predictive validity of the cyber hygiene inventory in explaining each of these was assessed by examining the correlations between the cognitive-behavioral constructs and each cyber hygiene dimension measured using the CHI. Table 6 presents the correlations between the three constructs and the CHI dimensions.

All the correlations were significant at the 0.05 level. The only exception was the relationship between email and messaging hygiene and cyber security self-efficacy that was significant at the 0.10 level (and significant at the one-tail level at 0.05, given that we expected cyber hygiene to lead to high self-efficacy). The only other exception was the relationship between online shopping related behaviors and authentication-credential hygiene, which came out to not be significant. This was likely because authentication hygiene, which taps into how default passwords are to be changed and how they should be created and stored, is largely unrelated to the users' carefulness in online shopping, where the measure focused on whether users assessed the domain name of the shopping site, the quality of the shopping sites pages, its images, colors, layout, and such.

4. Overall discussion

The research derived a conceptual definition of cyber hygiene and, using a series of studies, empirically developed the first-ever Cyber

Hygiene Inventory (CHI) measuring general Internet user cyber hygiene. The final 18 items, encompassing five dimensions of cyber hygiene, measure users' cognitive associations between different online actions and the extent to which the individual is aware of them.

The research makes the following three major contributions to the science of cyber security. First, the research identified the subdimensions of cyber hygiene. Until now, cyber hygiene has either been vaguely defined or been treated as a unidimensional facet of human cyber interaction. The research defined cyber hygiene as a second-order hierarchical construct and empirically found five sub-dimensions or latent factors of cyber hygiene, each covering a specific facet of online behavior. These factors, organized using the acronym SAFETY, encompass Storage and device hygiene, Transmission hygiene, Facebook and social media hygiene, Authentication and credential hygiene, and Email and messaging hygiene.

Second, the research developed the CHI, an 18-item summated scale measuring cyber hygiene. The scale can be used in its entirety in a survey or, alternatively, based on the focus of the research, individual subdimensions can be measured. The CHI, thus, makes it possible for practitioners, policy makers, and researchers to quantitatively assess the impact of each individual dimension on user cyber resilience, identify gaps, and provide pointed interventions. Furthermore, because the CHI is administered through a survey, it can be easily implemented at repeated intervals, which can help track the effectiveness of various interventions, thereby allowing for better resource allocation decisions overtime.

Additionally, the CHI can be applied to study different types of technology users almost anywhere in the world. This is because technology users all over the world face similar navigation, security, and accessibility experiences, thanks to the globalization of operating systems, platforms, devices, and services. Furthermore, the consumerization of technology along with new organizational developments such as work-from-home and bring-your-own-devices to work, has blurred the boundaries between the personal and professional work environments. For today's IT managers, what a user does at home on their devices and what malware they could port into the enterprise in consequence are even more of a concern than what they do on their better protected work devices. Hence, cyber hygiene expectations for the average consumer are also relevant for the employee in an organization, making the CHI relevant to organizational IT managers.

Finally, the research also demonstrated the impact of cyber hygiene on three facets of human cyber interaction: a self-belief about technology use, a critical facet of individual cognitive processing known to impact cyber resilience, and an online behavior that users all over the world regularly engage in. Cyber hygiene was shown to impact all of these significantly. This shows that cyber hygiene has an encompassing impact on what we think, how we think, and what we do online—all factors pivotal to cyber safety. On the one hand, the results establish the predictive validity of the CHI, while on the other, it also demonstrates the impact of cyber hygiene on secure human cyber interactions.

⁵ <https://www.statista.com/statistics/251666/number-of-digital-buyers-worldwide/>

Table 6
Correlations between CHI dimensions and cognitive-behavioral facets of human-cyber interaction.

CHI Dimension	Belief: cyber security self-efficacy	Cognitive: heuristic processing	Behavior: online shopping
F1: Storage and device hygiene	0.318**	0.123*	0.169**
F2: Transmission hygiene	0.199**	0.098*	0.139**
F3: Facebook and social media hygiene	0.140**	0.254**	0.141**
F4: Authentication and credential hygiene	0.168**	0.247**	−0.008
F5: Email and messaging hygiene	0.091 [#]	0.222**	0.168**

Note. **Correlation is significant at the 0.01 level (2-tailed); Correlation is significant at the 0.05 level (2-tailed); [#]Correlation is significant at the 0.10 level (2-tailed).

Despite its many advantages, there are also some limitations to be considered. For one, the inventory is new and this research marks the first step at developing and validating it. While the research established the intra and inter item to construct relationships, the generalizability or external validity of the inventory remains unknown. Two, in the present research, the CHI was tested on a broad swath of Internet users, but the extent to which it would apply in specific settings such as for employees working in a certain industry, a specific organization, or users performing a certain role, has yet to be determined. Finally, the inventory was mostly tested in a Southeast Asia nation. While the Internet is experienced similarly across the modern world, cultural and economic differences might still influence its use and, in consequence, the validity of the CHI might vary.

All of these weaknesses can be overcome by future research. While the present research established the internal validity of the measures, its external validity needs to be the focus of future research. This can be achieved by applying the inventory in different cultures, different organizational contexts, and, repeatedly, overtime. This could lead to case studies and best practice tips for IT managers in different parts of the world. Future research must also examine how CHI improves cyber resilience. This could be accomplished by administering the CHI and then conducting pen-test to assess whether users with high hygiene levels were less likely to fall for the test. Such research would not only establish the value of the CHI but also help identify what subdimensions of cyber hygiene actually matter from a resilience to different attacks point of view.

Overall, the research marks an important first step. While different groups have called for increased cyber hygiene, to-date there exists no measure or inventory for it, let alone any clarity about what it conceptually means or why it matters. The research defines cyber hygiene, identifies its underlying dimensions, develops an inventory for it, and shows why it matters. The research, thus, adds theoretically and empirically to the science of security.

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